# SakhiSathi Project Guide

Product concept, device teaser, companion app and technology plan

A working product and design guide for a women’s safety support system

Prepared 5 November 2026

| Project aim | Design principle |
| --- | --- |
| Help a person signal for support, share useful location information and practise safer responses. | Give the user control, make limitations visible and design for unreliable networks and stressful situations. |

This guide turns the current SakhiSathi concept into a product outline that can support a prototype, pitch, and requirements discussion. It describes a wearable or portable device, a companion mobile app, a community and learning area, and scenario-based practice. It is an early design document; final hardware, radio rules, emergency-service links, safeguarding practices and clinical/legal review must be confirmed for the pilot location.

## 1 Project overview

SakhiSathi is proposed as a personal safety system with two connected parts: a portable alert device and a mobile app. The device should make it quick to request help. The app should make setup, trusted contacts, location sharing, learning and practice understandable. The system should still communicate clearly when the phone is offline, permissions are missing or the device is not connected.

### Goals

- Offer a discreet and accessible way to trigger an SOS.
- Notify chosen trusted contacts and, where technically and legally available, connect the person to public emergency services.
- Use optional nearby community alerts only with consent, safeguards and clear limits.
- Teach practical, non-blaming safety steps and provide scenario practice without presenting a quiz score as a guarantee of safety.
- Protect sensitive location, incident and survivor-community data.

### Mind map

| SakhiSathi | Branches |
| --- | --- |
| People | User in distress; trusted contacts; trained community responders; moderators; emergency services |
| Device | SOS trigger; voice trigger; haptic/audio feedback; phone link; battery and connectivity status |

| | |
|---|---|
| App | Pairing; emergency workflow; live location; contacts; incident report; community; short lessons; scenario game |
| Infrastructure | Mobile network; GPS/location services; backend API; push/SMS fallback; moderation and audit tools |
| Trust and safety | Consent; privacy; false-alarm handling; abuse prevention; accessibility; testing; legal review |

# 2 Device concept and teaser

The teaser is a short product demonstration, not proof of emergency-service performance. It should show a believable sequence: the user activates the device, receives confirmation, the paired phone sends an alert and location to the selected recipients, and the app shows whether each delivery succeeded. Any RF beacon shown should be labelled as a local prototype alert, with its coverage and receiver requirements stated.

## Proposed user flow

| Step | What happens | What the user sees or hears |
|---|---|---|
| 1. Ready check | Device checks battery and Bluetooth connection to the phone. | App and device show ready, low battery, or disconnected. |
| 2. Trigger | User presses and holds the SOS control or uses a configured voice phrase. | Distinct vibration and/or tone confirms activation. A cancel window may be offered if safe and feasible. |
| 3. Alert | Phone app obtains current location and sends SOS to configured contacts and the backend. | Status shows sending, delivered, failed or location unavailable. |
| 4. Escalate | User can call 112. App may present a clear call action. Any automated integration requires formal approval and testing. | No claim that a radio signal alone contacts police. |
| 5. Nearby help | Optional receiver network or approved volunteer service may receive a coarse alert. | Only vetted recipients see limited information; no public broadcast of identity/location. |
| 6. Close | User ends alert or trusted contact helps close it. | App records a minimal event log and explains what data is retained. |

## Feature interpretation and feasibility

| Concept from brief | Recommended interpretation | Status / decision |
|---|---|---|
| Electric shock | Do not include an electric-shock function in the first product concept. It can injure the user or another person and creates major safety, liability and legal questions. Explore non-injurious attention features such as a loud alarm, flashing light or vibration. | Exclude from prototype pending specialist safety and legal review. |
| Radio-frequency alert to locals | A transmitter can only alert compatible receivers within range and on an authorized band. It cannot alert arbitrary phones, nearby people or police without a receiver network and service agreement. | Define the receiving devices, range, band, power, fallback and permissions. |
| “Tow mirror” | Meaning is unclear. This guide records it as an unconfirmed feature. If “two-way mirror” means camera detection, a mirror alone cannot reliably detect hidden cameras. If it means a rear-view/reflective aid, specify the use case. | Product owner to define before design. |
| SOS and live location | Phone uses GNSS/GPS and network location; the app shares location with selected contacts and indicates freshness/accuracy. | Core app capability, subject to permissions and connectivity. |
| Police reporting | Provide a user-initiated call to India’s 112 service or a clear handoff to the official 112 India app. Do not promise automatic police dispatch unless a formal integration exists. | Integration and operating process require confirmation. |

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|---|---|---|
| Voice module | On-device phrase detection may trigger the paired phone; use a narrow, configurable wake phrase and a physical SOS backup. Show activation feedback and a safe false-trigger cancellation method. | Prototype after privacy, language, noise and false-trigger tests. |

# 3 Technology architecture

Use the phone as the primary gateway to internet-connected services. A practical first prototype pairs a low-power button device with the phone over Bluetooth Low Energy (BLE). The phone can then attach its location and send messages using the available internet connection. This reduces the complexity of putting a cellular modem and GNSS receiver into the wearable, although it makes phone availability, permissions and battery important dependencies.

| Layer | Possible technology | Purpose and trade-offs |
|---|---|---|
| Device electronics | Microcontroller with BLE; physical SOS switch; vibration motor; buzzer/LED; battery and charging circuit | Low-power alert and status. Choose parts after industrial design, battery-life target and required certifications are known. |
| Device to phone | Bluetooth Low Energy, authenticated pairing | Short-range link to the owner's phone. BLE is not long-range community RF. Provide reconnect and lost-link status. |
| Voice activation | Local keyword spotting on a low-power chip or phone-side speech recognition | Local processing can reduce audio collection. Avoid always-uploaded audio; support language/accent testing and a physical fallback. |
| Location | Phone GNSS plus network-derived location, with timestamp and accuracy metadata | Phone location is usually the simplest first version. Device-side GNSS can be considered if the wearable must operate independently. |
| App | Android first or cross-platform Flutter/React Native; native modules for BLE, location, notifications and background execution | Choose based on team skill and device support. Confirm platform background-location and permission rules during implementation. |
| Backend | HTTPS API; managed database; authentication; queue/worker for notifications; role-based admin tools | Stores accounts, opt-in contacts, reports and community content. Keep emergency event data minimal and access-controlled. |
| Notifications | Push notification plus SMS/voice/call handoff fallback as appropriate | Delivery is not guaranteed. Show status; do not silently treat a queued alert as delivered. SMS provider costs and local requirements apply. |
| Optional local RF | Only after RF requirements are defined; possible receiver-based sub-GHz or mesh research subject to allowed bands, power, certification and regional rules | A local radio mesh needs compatible gateways/receivers and a coverage plan. It is not a replacement for cellular emergency services. |

## RF and radio decisions

- "Radio frequency" is a broad description, not a single technology. Select a protocol only after defining range, indoor/outdoor conditions, receiver density, message size, battery life, interference tolerance and operating region.
- BLE is suitable for the device-to-phone connection, not for alerting an arbitrary crowd.
- A mesh requires participating nodes that can forward messages. Define who owns and maintains those nodes, how relay is authenticated, and what happens when all nearby nodes are offline.
- For India, verify the selected radio module, operating band and product approval path with the current Wireless Planning and Coordination (WPC) requirements and a qualified compliance lab before field deployment.

# 4 Companion app specification

## Main areas

| Area | Key screens and behavior |
|---|---|
| Onboarding and setup | Plain-language explanation; device pairing; emergency contacts; permissions; test alert; location and notification settings; language and accessibility options. |
| Emergency home | Large SOS action; clear device/phone status; discreet option if relevant; visible route to call 112; test mode separated from real SOS. |
| Alert in progress | Status timeline; current location with update time/accuracy; trusted-contact delivery; stop/cancel control; call 112 action; low-network guidance. |
| Incident report | Optional structured notes, time and location; ability to save draft; consent before sharing; private by default; clear retention and deletion controls. |
| Community experiences | Moderated, opt-in survivor stories; anonymous or pseudonymous posting; reporting/blocking; no requirement to disclose graphic details or identifying information. |
| Calm and safety class | Short, accessible lessons on grounding, getting to a safer/public place when possible, contacting trusted people and emergency services, and preserving choice. Avoid blame or promises. |
| Scenario practice game | Fictional situations with choices and reflective feedback. No score should suggest a person is responsible for harm or guarantee an outcome. |
| Settings and privacy | Data export/deletion; trusted-contact management; consent and permissions; notification privacy; account security; language and accessibility settings. |

## Emergency app sequence

SOS press → device confirms → phone app reads paired-device event → app requests current location and timestamp → app sends alert to selected contacts/backend → app shows delivery results → user is offered a 112 call handoff → contacts can acknowledge → user closes the alert. If data service fails, show available alternatives such as SMS or phone call, subject to implementation and user settings. Avoid automatic police calls in the prototype.

## Community and moderation

- Default to pseudonymous participation and do not expose exact home, work or incident locations.
- Require clear consent before publishing, collecting, or sharing any story. Allow withdrawal and deletion requests.
- Provide reporting, blocking, content warnings and moderator escalation. Train moderators to avoid victim-blaming and to avoid offering professional counselling unless qualified.
- Have a response plan for imminent danger, self-harm disclosures, threats, illegal content and requests for help. State service hours and that peer posts are not monitored as an emergency channel.

## Calm and safety micro-class

Keep lessons brief (about two to five minutes), practical and optional. Possible modules: noticing immediate exits and public places; using a simple grounding exercise if it helps; reaching a trusted contact; making an emergency call; sharing location; and aftercare/support options. People respond differently under threat, so teach choices rather than one “correct” reaction. Ask a trauma-informed reviewer to review wording and illustrations.

## Scenario game design

| Component | Design |
|---|---|
| Scenario | Short fictional scene with context and accessible text/audio. Make clear that real events are unpredictable. |

| | |
|---|---|
| Choices | Offer several plausible options, including leaving, moving toward other people, contacting someone, using SOS or calling emergency services. |
| Feedback | Explain potential benefits and limitations of each option without grading the person’s worth or implying that one choice prevents assault. |
| Reflection | Ask what felt safe or difficult; allow replay or exit without penalty. |
| Evaluation | Evaluate comprehension of app controls and resource awareness, not real-world survival ability. Use aggregate anonymous analytics only with consent. |

# 5 Privacy, safety and reliability requirements

| Requirement | Minimum design direction |
|---|---|
| Consent and control | SOS contacts and sharing are chosen by the user; show who can see live location and when sharing ends. |
| Data minimization | Collect only what each feature needs. Avoid continuous tracking by default. Keep incident reports private unless the user chooses to submit/share. |
| Security | Encrypt network traffic; protect stored sensitive data; use secure device pairing; restrict staff access; log privileged access; plan security updates. |
| Reliability | Show battery, pairing, location accuracy/time, and delivery state. Provide physical SOS even if voice recognition fails. Test dead zones and interrupted phone service. |
| False triggers | Provide a tested cancellation flow that does not delay genuine requests; distinguish test alerts from live alerts. |
| Accessibility | Support large controls, screen readers, low literacy, multiple languages, hearing/vision needs and discreet use where feasible. |
| Abuse resistance | Protect users from stalkerware-like misuse, unauthorized pairing, contact coercion, location leaks and malicious community posts. |
| Emergency-service boundaries | Clearly state that SakhiSathi does not guarantee police dispatch or responder arrival. Make the official emergency route visible. |

# 6 Prototype roadmap and acceptance criteria

| Phase | Build | Exit criteria |
|---|---|---|
| 1. Requirements | Interview intended users and safety professionals; define “two-way mirror” question; choose pilot region and platform; map emergency and community workflows. | Approved feature list, privacy map, risk register and measurable pilot goals. |
| 2. App-only prototype | Clickable UX for SOS, contacts, location status, lessons, community mock and scenario game. | Users can find SOS, understand delivery status and configure contacts in moderated usability sessions. |
| 3. BLE device prototype | Physical SOS button, vibration feedback, BLE pairing and a test-mode app event. | Repeated trigger/pairing tests pass; low battery and disconnected-phone states are visible. |
| 4. Closed pilot | Small opt-in group; no shock function; no unapproved public RF; carefully monitored support processes. | Documented consent, support escalation, data retention, incident response and pilot evaluation. |
| 5. Controlled expansion | Consider RF receiver network and formal emergency-service integrations only after feasibility and compliance work. | Coverage survey, compliance evidence, service agreements and operational ownership are in place. |

### Pilot metrics

- Successful device-to-phone SOS trigger rate in controlled trials.
- Time to visible confirmation and time to contact delivery, separated by network condition.
- Percentage of alerts with location freshness/accuracy shown correctly.
- Setup completion and successful test alert rate.
- User understanding of what the app can and cannot do.
- Community report/moderation response time and privacy incidents.
- Lesson completion and scenario comprehension, without interpreting results as real-life safety outcomes.

# 7 Mind maps and system flows

### Device mind map

SakhiSathi device
├── Trigger: physical SOS / optional local voice phrase
├── Feedback: vibration / sound / light / status
├── Connection: BLE to paired phone
├── Phone actions: location / contact alert / 112 handoff
├── Optional network: authorized radio + compatible receivers
└── Foundations: battery / secure pairing / privacy / testing

### App mind map

SakhiSathi app
├── Emergency: SOS / live status / trusted contacts / 112 route
├── Device: pairing / battery / connectivity / test mode
├── Support: moderated community / incident reporting
├── Learning: calm and safety micro-lessons
├── Practice: scenario choices / reflective feedback
└── Controls: consent / privacy / language / accessibility / deletion

### End-to-end alert flow

| Trigger | Phone/device | Delivery | Recipient response |
|---|---|---|---|
| SOS button or configured voice phrase | Device signals paired phone; app confirms trigger and gathers location | Push/API to contacts; optional SMS/call; optional approved local receiver network | Recipient acknowledges; user sees status; user may call 112; event ends with user control |

# 8 Decisions to resolve

- Which user groups, languages and pilot region are first?
- What exactly does “tow mirror” mean?
- Is the first product a wearable, keychain, phone accessory or another form?
- Should voice activation run locally on-device, on the phone, or be omitted in the MVP?
- Who receives the local RF alert, through what hardware, and who maintains coverage?
- Which emergency contacts are notified and what is the opt-in policy for nearby volunteers?
- What incident data is retained, for how long, and who can access it?
- Which platform and team skills determine Android-only versus cross-platform development?

# 9 Reference points

These sources ground the emergency-service and radio-compliance decisions. Requirements change; confirm current rules and local arrangements before a pilot.

- Emergency Response Support System in India, 112.gov.in — https://112.gov.in/
- ERSS overview and 112 India app, 112.gov.in — https://112.gov.in/about
- Ministry of Women and Child Development women’s helpline information — https://wcd.gov.in/public/women/help
- Department of Telecommunications, Wireless Planning and Coordination Wing — https://dot.gov.in/wireless-planning-coordination

Product note: India’s official ERSS identifies 112 as the pan-India emergency number. The government describes the 112 India app as an emergency route that can send location; the 112 site also describes SHOUT alerts to registered volunteers for women and children. SakhiSathi should link or hand off to official routes unless a direct integration is formally authorized and operationally supported.